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Data Strategy

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Overview

- Introduction
- Why study “business strategy” / what is “business strategy”?
  - Focus on the world of VCs
- History of Business Strategy (1900-Present)
- A few take-aways and predictions

About me (Academically)

- PhD at UCLA in Management (2012)
- Masters in Economics at UC Davis (2007)
- BA in Applied Math/Statistics at UC Berkeley (2002)
- Research Area(s):
  - Financial Markets
  - Marketing
  - Design of Experiments
  - Applied Big Data ”Stuff”



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About me (Professionally)

- University of Chicago (2022-Present)
  - Director, Data Science Clinic
- Other current stuff:
  - VC Due Diligence, Investor, etc.
- Director of Backend Engineering & Data Science at The Meta (Not FB, Kovaak) 2020-2022
- Assistant Professor of Data Science at USF (2014-2020)
  - Director of Industry-Academic Partnership
- Director of Analytics at Sega (2014-2015)
- Director of Analytics, Backend Engineering and User Acquisition at TinyCo (2011-2014)
- Senior Consultant Bates White (2002-2006)



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Some Games I've Worked on



















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Business Strat

- Big Topic. You can get a PhD in it.
- Obviously not going to cover everything.
- What we will cover:
  - Motivation
  - Generational waves of beliefs and frameworks
  - Focused on the DATA side.
- Why do this?
  - This is how many VCs and investors think (and talk) about investments.
  - This is where the pool of questions will come from



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Business Strat

- The other WHY:
  - Knowing these investment thesis and why they no longer make sense will allow you to avoid mistakes:
    - When raising money
    - When planning
    - Allow you to avoid bad investors
  - How to respond to some common business questions (to anyone, not only investors)
  - Finally – I find it interesting. HA!



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Business Strat: Motivation

- Many definitions:
  - Optimizing “long-term” company goals
  - Optimize allocation of company resources
  - Optimize course of actions
  - Optimize decision making
  - Optimize how to compete
  - Optimize the policy to get to goals
  - …

Business Strat: Motivation

- These questions all revolve around one thing:
  - “Why do some companies or industries make money and others do not?”
  - And… “how do we make sure it is our company?”
- This is (obviously) an incredibly important question and it has a long history.

Business Strat: Motivation

- Disclaimer:

Before jumping in – the next section is a very biased and incomplete history which allows us to quickly jump to the point where we can talk about issues related to data *today.*

If you are interested in a real history of corporate strategy, there are many great articles and books. This is not one.



Business Strategy

History, part I

A very opinionated set of stylized facts

Up to 1999

Business Strat: History

- Two Core Components (late 1800's, early 1900's)
  - Efficient Market Hypothesis
    - Asset prices reflect all available information
  - Zero Profit condition of capitalism
    - Companies will enter an industry until profit goes to zero
    - MC = MR for profit max, etc.
- General belief until around the great depression

Business Strat: History

- 1930-1950: “What is competition?”
  - Imperfect/limited competition could yield sustained profit
  - *Monopoly* profitability:
    - Industrial organizational plays a role in long-term profitability.
    - Monopoly can be sustained by their actions.
- 1950-1970s: Execution Era
  - Even within relatively perfectly competitive industries you still see some companies having sustained profit.
  - Some companies are just “better than others” at doing their job.
  - A lot of biz strat at this time focused on “how to execute” and “how to organize”
    - GE “teams you can put your arms around”
    - Centralization / Decentralization / how to organize and function

Business Strategy

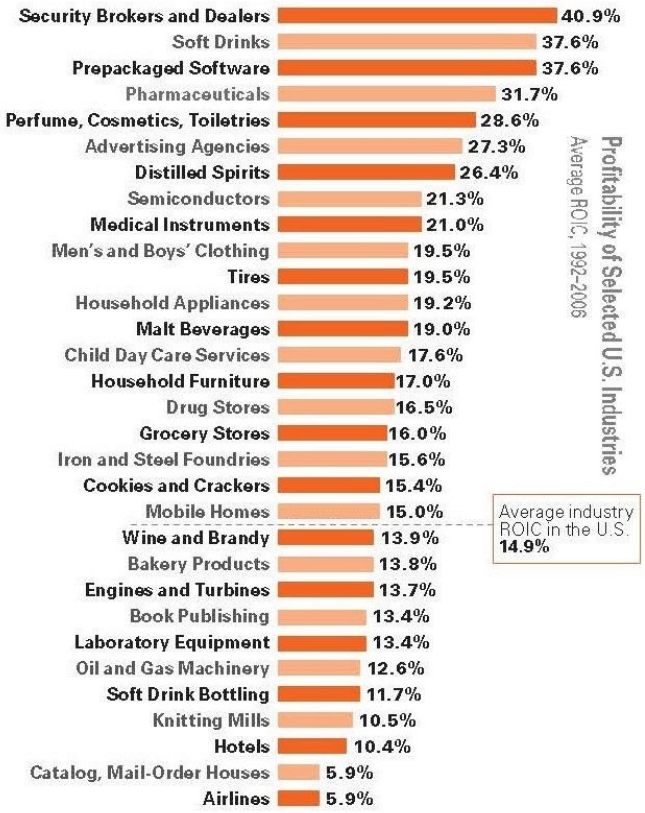
History, part II

A very opinionated set of stylized facts

1979-1999

Business Strat: History

- We then get to the late 1970s
- This chart shows profitability by *industry*
- It doesn't make sense given the previous analysis:
  - Companies can *execute* better, but entire industries should not
- Companies *should* enter these markets, drive down profitability
- *Industries* should not be profitable



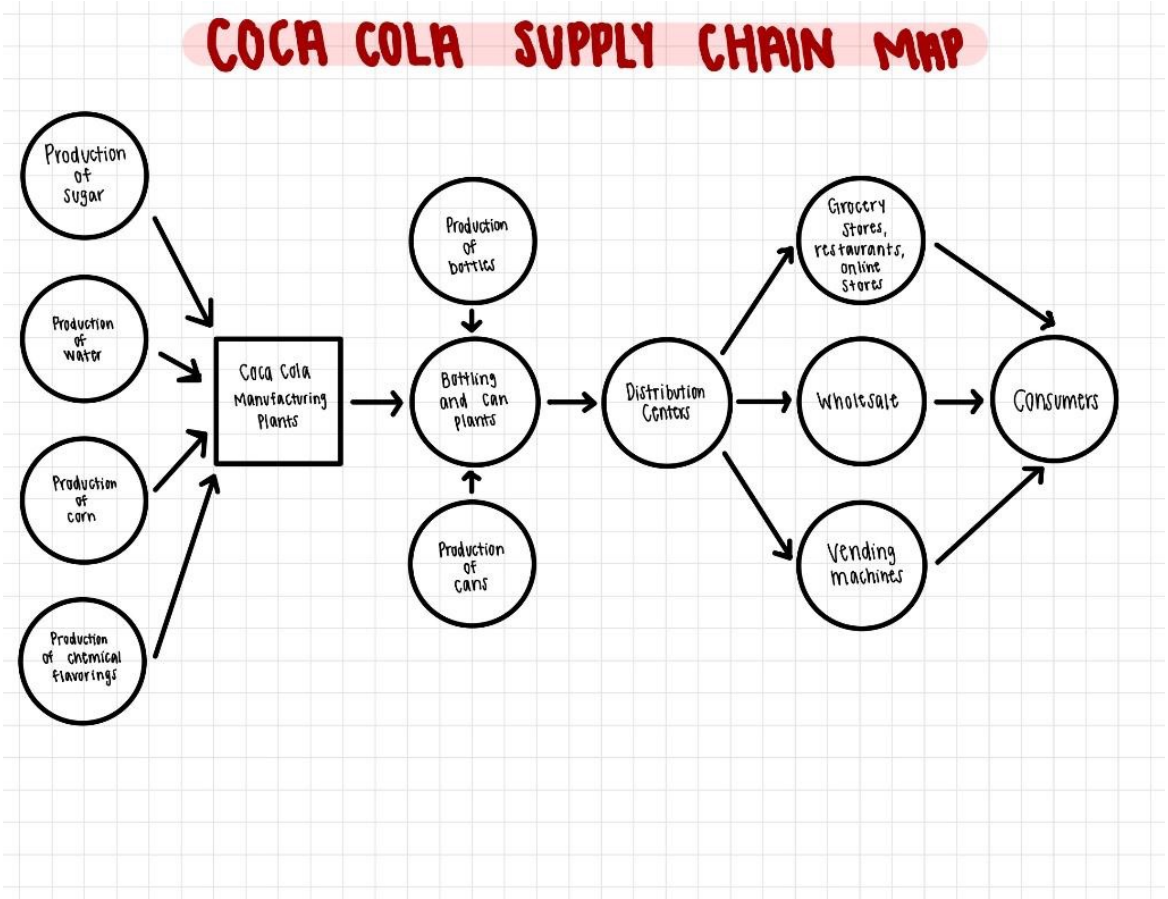


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Business Strat: History

- Key insight:
  - Consumer prices are not fully set by supply/demand activities, but also by a customer's **willingness to pay** a specific price.
  - That **willingness to pay,** assuming it is larger than the cost of producing the good/service will generate surplus.
  - There are ***structural*** forces which determine who then captures that surplus.

Business Strat: History



- Coca-Cola costs $2.50 at the store.
- The total cost to produce is $2.10.
- Why does Coca-Cola keep nearly all the surplus?



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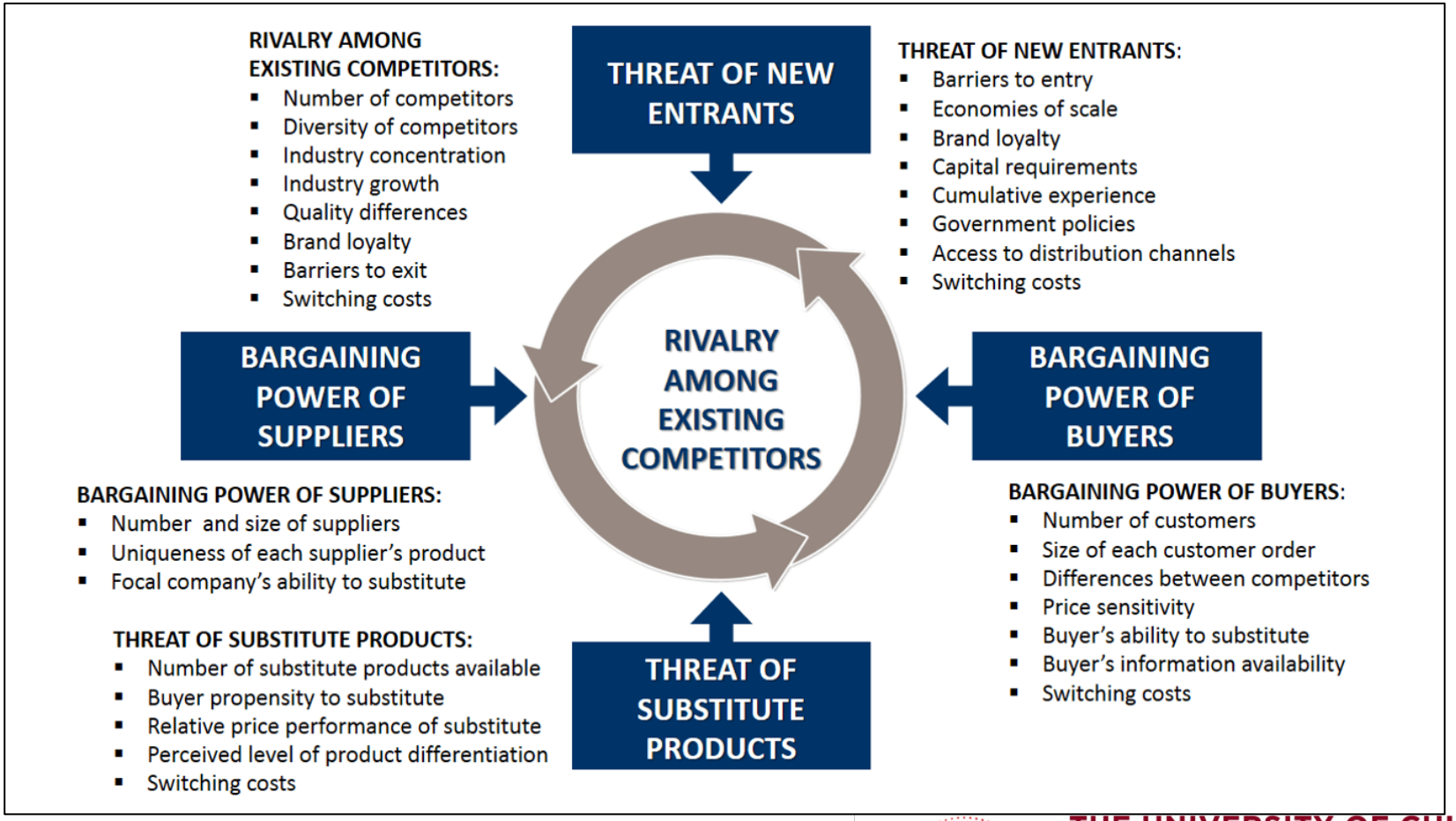
Business Strat: History

- Michael Porter (HBS)
- Put forth a structured way of thinking about “forces” that act on **industries.**
- Porters Five Forces Model
- Determines who (industry) captures surplus.
- Published in 1979



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Business Strat: History



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Business Strat: History

- As the forces intensity increases => less long-term profitability.
- 40 years of MBA students study Porter's model and are required to do a "Five Forces Analysis" on an industry
- To this day people continue to use this framework. And I strongly recommend all of you to take the time to apply it to your company – if only as an organizing exercise.
- It is a very useful checklist



Business Strat: History

- 1979-1995 nothing (big) changes
- Common Criticisms raised:
  - What is an industry?
  - Execution still matters (Boeing vs. Airbus)
  - Not all forces are equal – how can we quantify “high” vs. “low”?
  - (To some) model understates background issues:
    - Government, international rules, etc.
  - Not predictive – more of a snapshot or a look backward.
  - Doesn’t provide remedies.

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History, part III: Modern History

A very opinionated set of stylized facts

1995-2022

Dot-com 1.0 (1995-1999)

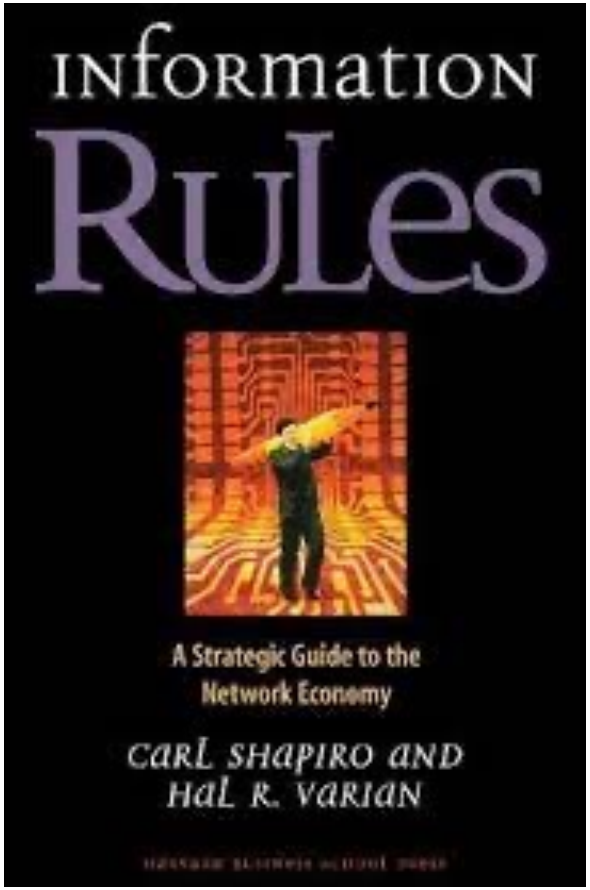
- Late 90's wave of .com companies "break" a lot of what is going on.
- Tons of confusion.
- Yahoo, Napster, paypal, etc.
- Why were they worth so much?
  - They didn't make a "good" in the classic sense
  - The "service" they provided didn't fit into the *classic* service economics (law firms, accountants, consulting
- In late 90's people were talking about the "New Economics" and "New Strategies"
  - Business news at this time is *wild*
- The first "data is exploding" era
- What is happening?!?!



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Information Rules Era: 1999-2013

- Varian / Shapiro wrote ”information rules”
- Berkeley Professors:
  - Shapiro: Biz Strat Professor at UC Berkeley
  - Hal Varian: Emeritus professor at UC Berkeley, chief economist at Google
- They are “hard math” people



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Information Rules Era: 1999-2013

- Fundamental Argument:
  - The same business strategies that existed before are still useful and valuable. However, and this is a big BUT, their relative values have changed.
  - Example:
    - Having a good inventory system = better prediction of future demand = better able to negotiate contracts with suppliers = profits.
    - Why not before? Information is now way, way cheaper.
  - There is a new *class* of "good" that must be treated separately: ***information good.***
    - Music, writing, computer programs and models.
    - High fixed cost, but **zero marginal cost.**
    - If MC = 0, what should it be priced at? (…value to consumer)



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Information Rules Era: 1999-2013

- Impact:
  - Product differentiation is WAY more important
    - Need to tie in that differentiation with pricing strategy
    - Differential Pricing
  - Right managements: how do you maximize your IP to *manage* its value – rather than just trying to protect it.
    - Napster / music industry
  - Lock-in must be managed differentially
  - Positive Feedback and network externality management.
    - Almost all information goods have some type of network externality
  - Standards and Alliances:
    - How do you manage standards with information products?

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Information Rules Era: 1999-2013

- Reshuffling of priorities within traditional industries to align with the informational aspects of the above:
  - Differential pricing started to become ubiquitous.
  - “Pricing teams” / “Price Discrimination” / etc. are all now accepted parts of the economy

Information Rules Era: 1999-2013

- Key take-aways:
  - Treat information goods as a separate *type* of product
  - All the Porter style forces are still important, but their relative values are now changed.
  - Information (on our customers, inventory, etc.) provides additional opportunity:
    - Price discriminate
    - Lock-in
    - Identify new opportunities
    - Better manage our business
    - Etc.
  - Things that used to be “off the table” due to cost are now open to optimization! (Execution)



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Data is the new oil: 2013-2017

- If information powers potential biz improvements…
- 2013 = “FAANG” Coined
- “data has intrinsic value”
- Flywheel:
  - If I have data => biz is better (profits) => generates more data => biz is even better (profits) => etc.
- Cover = 2017



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Data is the new oil: 2013-2017

- For the FAANG companies this proved (relatively) true:
  - Google/ Facebook => more info => better ad targeting => more profits
  - Amazon => more info => better recommendations => more profit
  - Netflix => more info => create shows to target => more profit
  - What about apple?



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Data is the new oil: 2013-2017

- Tons of VC investment in the data side.
- “Applications” leveraging data as a core part of their business
  - Airbnb / lyft / uber
- “infrastructure” used to help companies use data
  - Mixpanel, mongdb, etc.
- Turns out data is NOT the new oil:
  - Transferability of data
  - Short term value of data:
    - Data is closer to a liability than an asset.



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The (empty) promise of Data Moats: 2017-2020

- “Moats”… sigh.
  - *Sounds a lot like “forces”*
- “systems thinking” (Jerry Chen had a good summary)
- Systems of Records:
  - How data is recorded and accessed
  - Defensible ”moats” due to pure network effects:
    - Salesforce, Workday, peoplesoft, SAP and Oracle
    - Provide a foundation upon which other companies build things, but end up being the dominant player in a vertically isolated space.



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The (empty) Promise of Data Moats: 2017-2020

- Systems of Engagement:
  - How we interact with that underlying system of record
  - Moats are based on network effects and usability:
    - Each successive generation knows how to use system X, so that is what they use:
    - WeChat, Slack, etc.
- Systems of Intelligence:
  - We can use the two systems “below” to create models and feed those results back into engagement in order to create value.
  - Virtuous cycle of model performance:
    - Model retention best => people will buy their product. They will have the best data => create better models (flywheel v2 @ the model level)



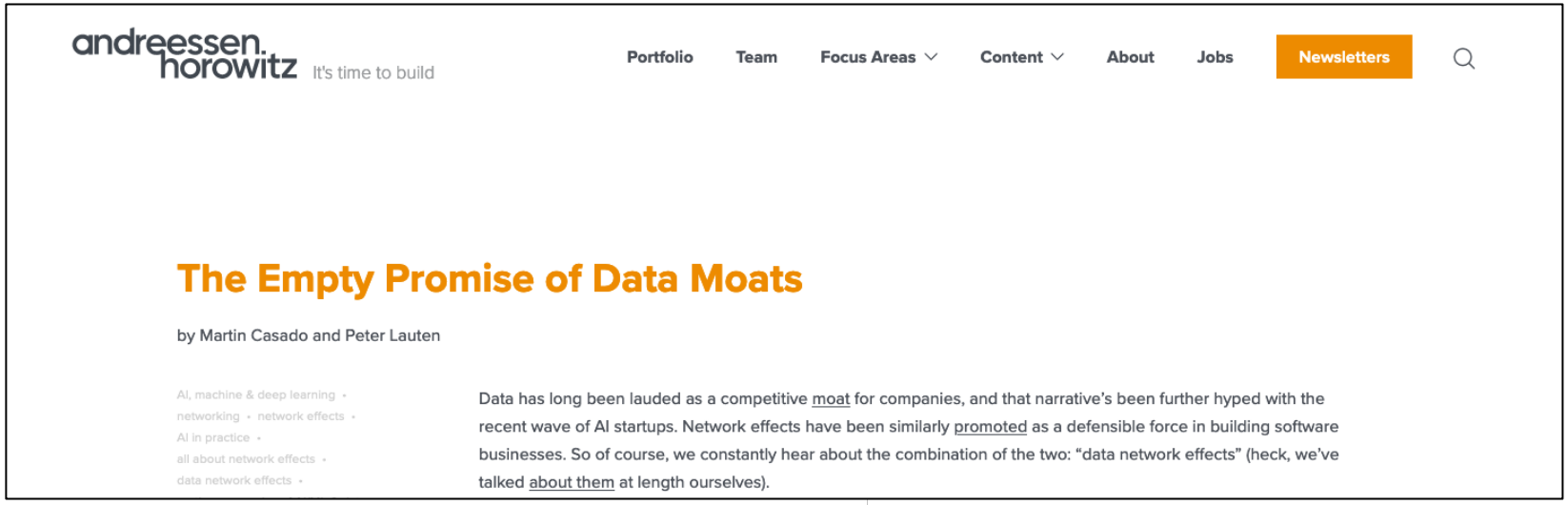
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The (empty) Promise of Data Moats: 2017-2020

- In essence:
  - *data* isn't a moat,
  - Algorithms are a moat.
- (The king is dead, long live the king)

The (empty) Promise of Data Moats: 2017-2020

- 2020 brought about the death (IMO) of much of the “data is special talk”
- “The Empty Promise of Data Moats”



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The (empty) Promise of Data Moats: 2017-2020

- Why?
  - Marginal Information Collection costs skyrocket
  - Marginal information value also decreases
  - Stale data
  - Long tail of “data use” has limited ROI.
    - Think of a chat bot, if I can get through 80% of the questions people ask correctly, the ROI of moving from 80->90 is limited.
- Money quote of the article:

Greater long-term defensibility is more likely to come from ***packaging differentiated technology***

(sounds a lot like Varian / Shapiro)



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The (empty) Promise of Data Moats: 2017-2020

- Consulting vs. Technology companies:
  - ”AI” Companies were behaving more like consulting companies from an accounting perspective.
  - Traditional software companies: “Information Goods”
    - Low marginal cost, high fixed
  - AI companies:
    - High marginal costs, low fixed
  - Why?
    - Training models (and running them)
    - Human Intervention & Human in the loop
    - Cloud Costs
    - Onboarding costs
- Service companies like this => not why VCs do this.



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The (empty) Promise of Data Moats: 2017-2020

- Where does that leave us if we are an “AI” biz:
  - Lower custom costs on AI businesses to perform more like a traditional tech company
  - Price in variable costs in order to increase profitability
  - Embrace Services, but price them out



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2020-2022: Blockchain/Crypto

- Huge amounts of capital raised (a16z 4.6billion fund), but deployment has slowed significantly
- A lot of TBD here.
  - Cynic: There is no here here. / web3isgoinggreat
  - Optimistic: Crypto-twitter
  - Truth: somewhere in the middle

2022-2023: LLM & Applications

- This area is changing FAST.
- Tough to predict who is going to be a winner. Some possibilities:
  - OpenAI collect much of the surplus
  - Infrastructure to build internal, company specific versions collect surplus
  - Applications capture this moat (jasper.ai for example)
- Third seems doubtful to me, but what do I know?



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Where does that leave us

- I've not heard of any current prevalent "thesis" on data and data value.
- That doesn't mean you can't sell it as part of your business, but it means you must position it accordingly.
- I suspect that selling data, as part of your company in an "information goods" style package will be met well by investors.
- Learn from the past -- if you are going to try to sell based on one of these frameworks ideas, make sure you have concrete answers to why it is different.



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Appendix: Fun Example

- Tomasz Tunguz – well-known investor
- Publishes a well-read newsletter / blog
- Says that “Usage will be a moat” for generative AI startups…



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Appendix: Fun Example

Usage as the Moat in AI

As generative AI captivates Startupland, startups will do what they have always done: integrate new technology to build transformative businesses.

Incumbents have seized the moment with Microsoft, Adobe, & others integrating generative AI into their products quickest.

In response, startups must develop moats to stake out their market. What are these moats?

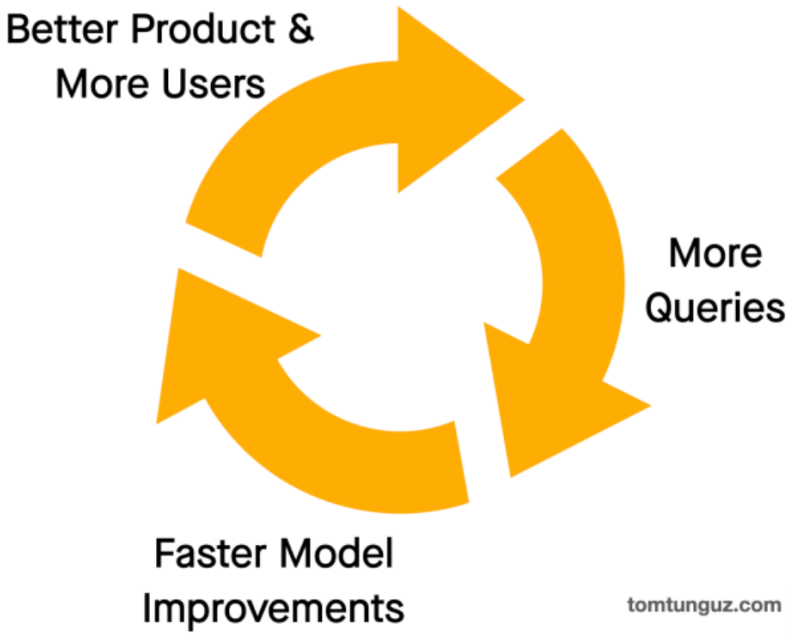
At the moment, capital & technical expertise create competitive advantage. Models require millions of dollars & technical expertise to deploy: document chunking, vectorization, **prompt-tuning** or **plugins** for better accuracy & breadth.

But in the long-term, usage will be the enduring moat.



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Appendix: Fun Example



That process spins a flywheel. More queries -> more diagonostic data to improve the model -> a better product with more users.

Appendix: Fun Example

- Argument looks a lot (to me) like the previous algorithmic moats we spoke about
- Which didn't work because models and technology changed to quickly for stale/marginal data to be valuable to current questions.
- Will the same happen here?
  - Future will tell!
- I'd say so, but what do I know?
- If I were trying to "sell this moat" – I'd make sure to be able to easily describe how this is different than previous algorithmic moats.



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The End



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